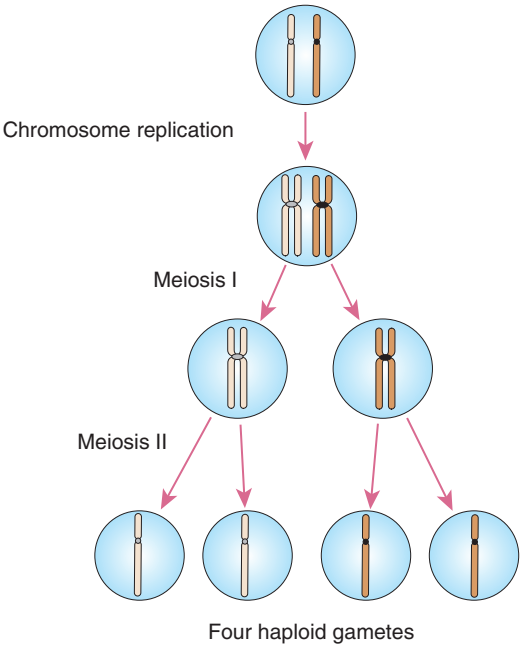


Figure 2-13 A simplified representation of the essential steps in meiosis, consisting of one round of DNA replication followed by two rounds of chromosome segregation, meiosis I and meiosis II.

called the **synaptonemal complex,** which is essential to the process of recombination. After synapsis is complete, meiotic crossing over takes place during **pachytene,** after which the synaptonemal complex breaks down.

Metaphase I begins, as in mitosis, when the nuclear membrane disappears. A spindle forms, and the paired chromosomes align themselves on the equatorial plane with their centromeres oriented toward different poles (see Fig. 2-14).

Anaphase of meiosis I again differs substantially from the corresponding stage of mitosis. Here, it is the two members of each bivalent that move apart, not the sister chromatids (contrast Fig. 2-14 with Fig. 2-9). The homologous centromeres (with their attached sister

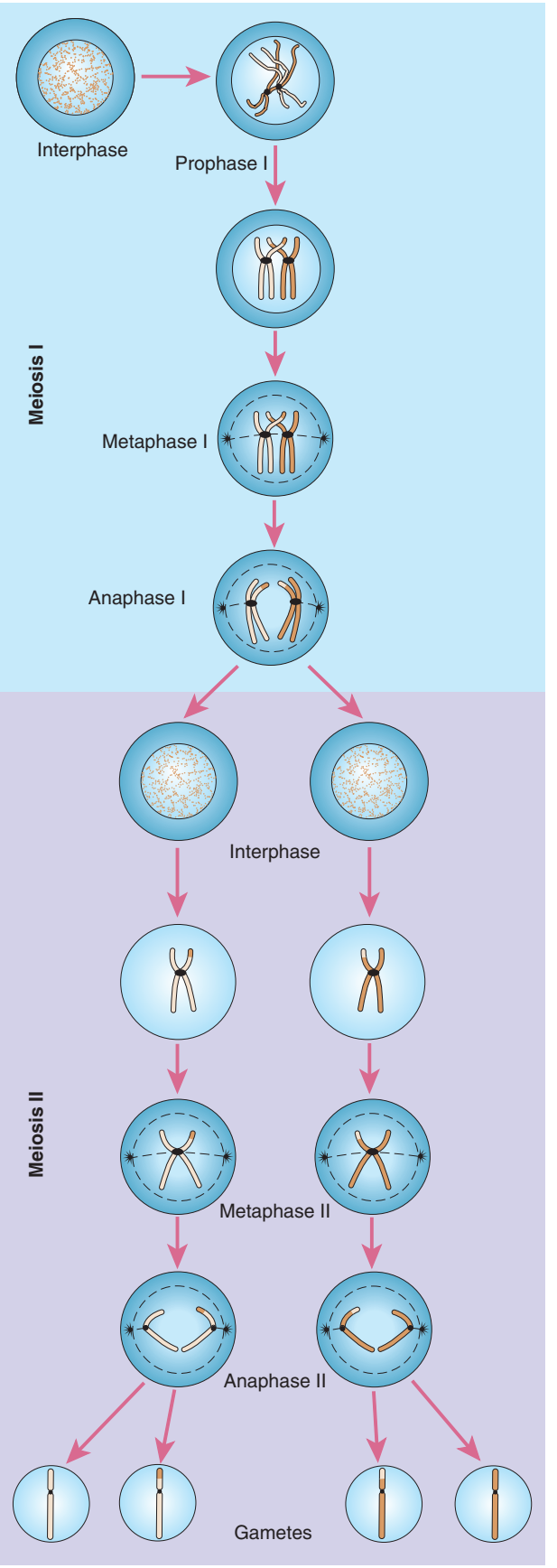


Figure 2-14 **Meiosis and its consequences.** A single chromosome pair and a single crossover are shown, leading to formation of four distinct gametes. The chromosomes replicate during interphase and begin to condense as the cell enters prophase of meiosis I. In meiosis I, the chromosomes synapse and recombine. A crossover is visible as the homologues align at metaphase I, with the centromeres oriented toward opposite poles. In anaphase I, the exchange of DNA between the homologues is apparent as the chromosomes are pulled to opposite poles. After completion of meiosis I and cytokinesis, meiosis II proceeds with a mitosis-like division. The sister kinetochores separate and move to opposite poles in anaphase II, yielding four haploid products.